

n8n



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Agenda

Was Sie erwarten können



1. Introduction
2. Node Types
3. Workflows and Credentials
4. Workflow “Getting Started”
5. Workflow “AI Agent”
6. Workflow “Competitor Analysis”
7. Workflow “RAG in n8n (local)”
8. Workflow “AI News for HackerNews”
9. Workflow with Webhooks
10. Workflows via MCP



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Introduction

- **Workflow Automation Tool**
 - open-source platform designed to help automate tasks and connect various services together
- **Low-Code/No-Code Approach**
 - primarily uses a visual, node-based interface,
 - allows users to build complex workflows by dragging, dropping, and connecting blocks (nodes) without needing to write extensive code
- **Extensible and Self-Hostable**
 - can be run on your own server (self-hosted), giving you complete control over your data and infrastructure.
 - also supports custom code nodes for advanced functionality.
- **Node-Based Architecture:** A "node" represents a specific app, an action, or a piece of logic
- "workflow" is a series of interconnected nodes



- businesses use dozens of separate tools (CRM, databases, messaging, file storage)
- n8n acts as the central integration layer
- breaks down data silos.
- automates repetitive Tasks - takes over manual, time-consuming, and error-prone tasks
- Example
 - Automatically moving new contacts from a web form (e.g., Google Forms) into a CRM (e.g., Salesforce) and sending a Slack notification.
- Enables custom, complex logic - allows for conditional logic, data transformation, and custom scripts to handle highly specific business process requirements
- Ensure Data Privacy and Security

- **Significant Time & Cost Savings:** By automating business processes, organizations reduce the operational time and labor costs associated with manual data entry, processing, and application switching.
- **Increased Efficiency and Accuracy:** Automated workflows execute tasks consistently and flawlessly, eliminating human error and speeding up critical processes.
- **Flexibility and Customization (Open Source):** The community-driven nature means it has a vast library of nodes (currently over 350) for various apps. If a node doesn't exist, users can build their own or use the Function node for custom JavaScript code.
- **Scalability:** Workflows are designed to handle growing volumes of data and can be easily duplicated, modified, and scaled across different departments.
- **Empowers Non-Developers (Citizen Integrators):** The visual interface allows non-technical users to quickly understand, build, and maintain powerful integrations without relying solely on IT/development teams.



Node Types

Trigger Nodes

- Initiate a workflow
- typically based on a schedule, a webhook, or an event in an external service

Action Nodes

- Perform tasks
- connect to external APIs and services
- manipulate data (e.g., HTTP Request, Send Email, Airtable, Google Sheets).
- Regular Nodes

AI Nodes

- Nodes that integrate AI services and models for intelligent decisions or content generation
- e.g., nodes for OpenAI, Claude, or specific AI Chains/Agents

AI



Build autonomous agents, summarize or search documents, etc.

Action in an app



Do something in an app or service like Google Sheets, Telegram or Notion

Data transformation



Manipulate, filter or convert data

Flow



Branch, merge or loop the flow, etc.

Core



Run code, make HTTP requests, set webhooks, etc.

Human in the loop



Wait for approval or human input before continuing

Workflows and Credentials

Editor

Executions

Evaluations

Readme

Load your data into a vector database with the Load Data flow, and then use your data as chat context with the Retriever flow.

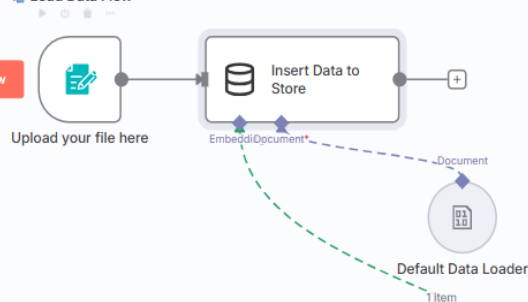
Quick start

1. Click on the [Execute workflow](#) button to run the Load Data flow.
2. Click on [Open Chat](#) button to run the Retriever flow. Then ask a question about content from your document(s)

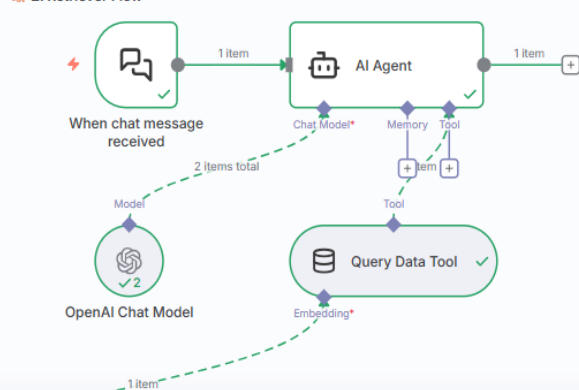
For more info, check [our docs on RAG in n8n](#).

Execute workflow

Load Data Flow



2. Retriever Flow



Embeddings

The Insert and Retrieve operation use the same embedding node.

This is to ensure that they are using the exact same embeddings and settings.

Different embeddings might not work at all, or have unintended consequences.

Execute workflow

Hide chat



Google Docs
Google Sheets
Google Calendar
Google Gmail
...

1. Create Google Cloud Project
<https://console.cloud.google.com/>
2. Enable APIs
3. OAuth Consent Screen
4. Redirect URL
5. Publish App

API APIs und Dienste

Youtube video: <https://www.youtube.com/watch?v=FBGtpWMTppw>



Workflow „Getting Started“

Triggers



When clicking 'Execute workflow'



Schedule Trigger



Typeform Trigger

Form ID:

Storage



Append row in sheet
append: sheet



Insert row



Webhook1



Respond to Webhook

Data Processing



Split Out



Aggregate



If

true
false



Switch
mode: Rules



Code in Python (Beta)



Input 1
Input 2
Merge
append



Workflow „Simple Agent“

Try It Out!

Launch your first AI agent—a chatbot that uses tools to fetch live info, send emails, and automate tasks.

To get started:

1. **Connect Gemini** (see red sticky note below)
2. Click the **Open chat** button and try asking:
 - "What's the weather in Paris?"
 - "Get me the latest tech news."
 - "Give me ideas for n8n AI agents."

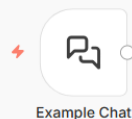
Questions or Feedback?

For feedback, coaching, built-for-you workflows or any questions, use my unified AI-powered contact form.

[Get In Touch Here](#)

Happy Automating! —Lucas Peyrin

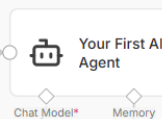
Later, activate this workflow and share the public chat URL to let others talk to your AI Agent!



Your AI agent can:

1. **Receive** messages from the chat
2. **Select** the right tools (e.g., weather, news, email)
3. **Respond** with live, helpful answers

Open the **AI agent** node and edit the **System Message** to adjust your agent's thinking, behavior, and replies.



Chat Model* Memory Tool

Model



Connect Gemini

1. In **Google AI Studio** click **"Create API key in new project"** and copy it.
2. Open the **Connect Gemini** node:
 - **Select Credential** → **Create New**
 - Paste into **API Key** and **Save**

Memory



Conversation
Memory

This node helps your agent remember the last few messages to stay on topic.



Get Weather



Get News

These tools let your agent access real-world data or take actions. Add more to expand its abilities!

Click the **+** under the agent's Tool input to add:

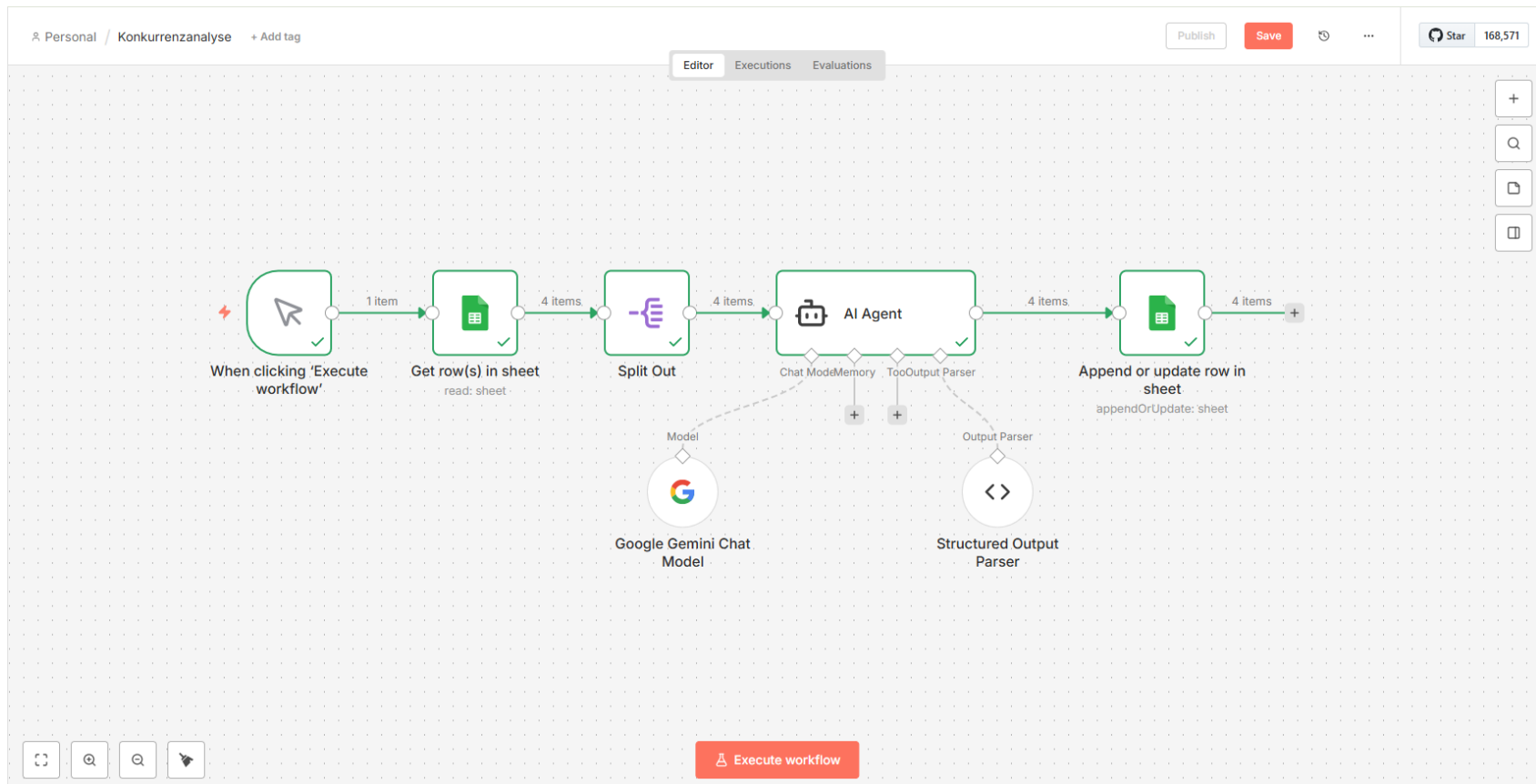
- Google Calendar (Get Upcoming Events)
- Gmail (Send an Email) (Gmail)



Open chat

Workflow „Competitor Analysis“

Workflow „Competitor Analysis“





Initial Situation

	A	B	C
1	Firmenname	Firmensitz	Firmenzweck
2	adidas		
3	airbus		
4	allianz		
5	basf		

- List of competitor companies
- We need to research information for all of these companies.



Final Result

	A	B	C
1	Firmenname	Firmensitz	Firmenzweck
2	adidas	Herzogenaurach	Adidas ist ein welt
3	airbus	Toulouse, Frank	Airbus ist ein welt
4	allianz	München, Deuts	Das Anbieten von
5	basf	Ludwigshafen a	BASF ist ein mult
6			

Workflow „RAG in n8n (local)“

Readme

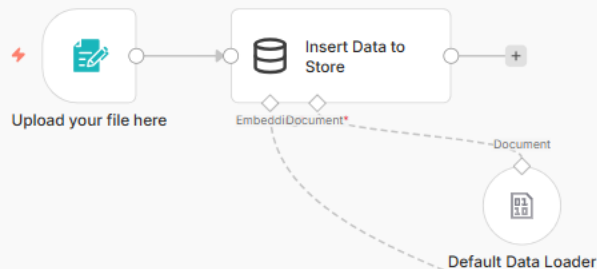
Load your data into a vector database with the **Load Data** flow, and then use your data as chat context with the **Retriever** flow.

Quick start

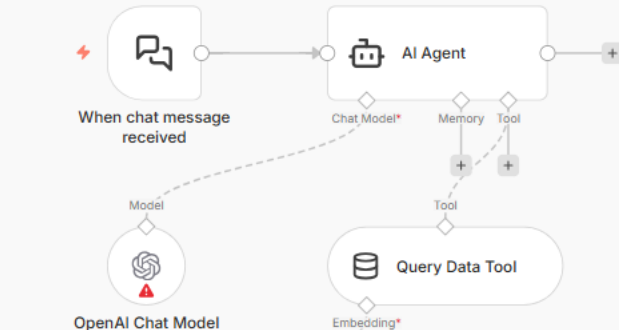
1. Click on the **Execute Workflow** button to run the **Load Data** flow.
2. Click on **Open Chat** button to run the **Retriever** flow. Then ask a question about content from your document(s)

For more info, check [our docs on RAG in n8n](#).

Load Data Flow



2. Retriever Flow



Embeddings

The Insert and Retrieve operation use the same embedding node.

This is to ensure that they are using the **exact same embeddings and settings**.

Different embeddings might not work at all, or have unintended consequences.

Embeddings OpenAI

Workflow

„AI News from HackerNews“

Workflow „AI News from Hacker News“

Personal / AI News Digest from Hacker News

Daily AI News Digest from Hacker News

What This Template Does

Fetches recent AI stories from Hacker News, filters for today, scrapes each article, summarizes with GPT into snippets, aggregates into HTML, and emails a styled daily digest.

Prerequisites

- n8n instance
- OpenAI API access
- SMTP server (e.g., Zoho, Gmail)

Required Credentials

OpenAI API

- Get key from platform.openai.com/api-keys
- Add to n8n as OpenAI credential
- Assign to "GPT 5 pro" node

SMTP Setup

- Configure in n8n credentials (host, port, user/pass)
- Assign to "Send Email Digest" node
- Update fromEmail/toEmail fields

Configuration Steps

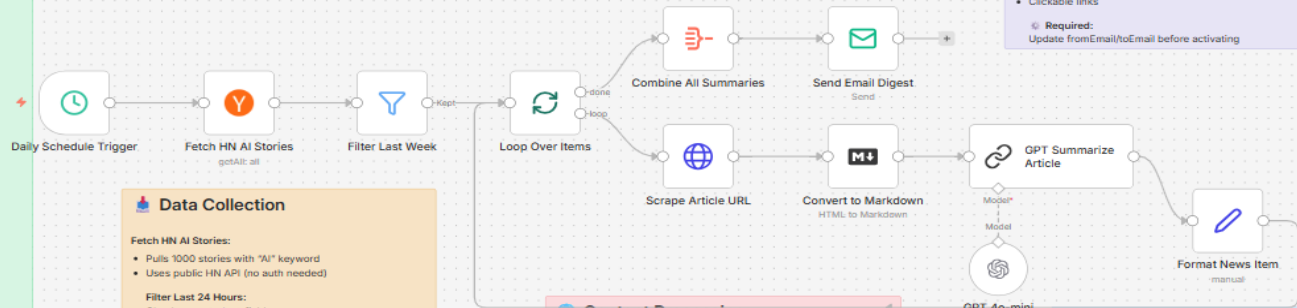
- Import JSON into n8n
- Assign OpenAI and SMTP credentials
- Update fromEmail/toEmail in "Send Email Digest" node
- Set schedule in "Daily Schedule Trigger" (e.g., daily at 8 AM)
- Activate workflow

Use Cases

- Personal AI news briefing
- Team knowledge sharing
- Content curation for blogs

Troubleshooting

- No stories: Check keyword filter; HN API limits
- Scraping fails: Add proxy if blocked
- Email not sending: Verify SMTP settings
- Summaries poor: Tweak GPT prompt



Data Collection

Fetch HN AI Stories:

- Pulls 1000 stories with "AI" keyword
- Uses public HN API (no auth needed)

Filter Last 24 Hours:

- Checks `created_at` field
- Keeps only yesterday/today posts
- Uses `$now().minus({ days: 1 })`

Customization:
Change keyword filter or date range here

Content Processing

Loop Over Items:

- Processes each story sequentially
- Prevents rate limiting

Scrape Article URL:

- Fetches content from source URL
- May need proxy for blocked sites

Convert to Markdown:

- Cleans HTML to text
- Formats for LLM input

Troubleshooting:
If scraping fails, check URL accessibility

Email Delivery

Send Email Digest:

- Styled HTML template
- Mobile-responsive design
- Inline CSS for compatibility

Template Features:

- Clean news item layout
- Branded header/footer
- Clickable links

Required:
Update fromEmail/toEmail before activating

AI Summarization

GPT Summarize Article:

- Generates heading + 2 sentences
- Extracts key AI insights
- Includes link when relevant

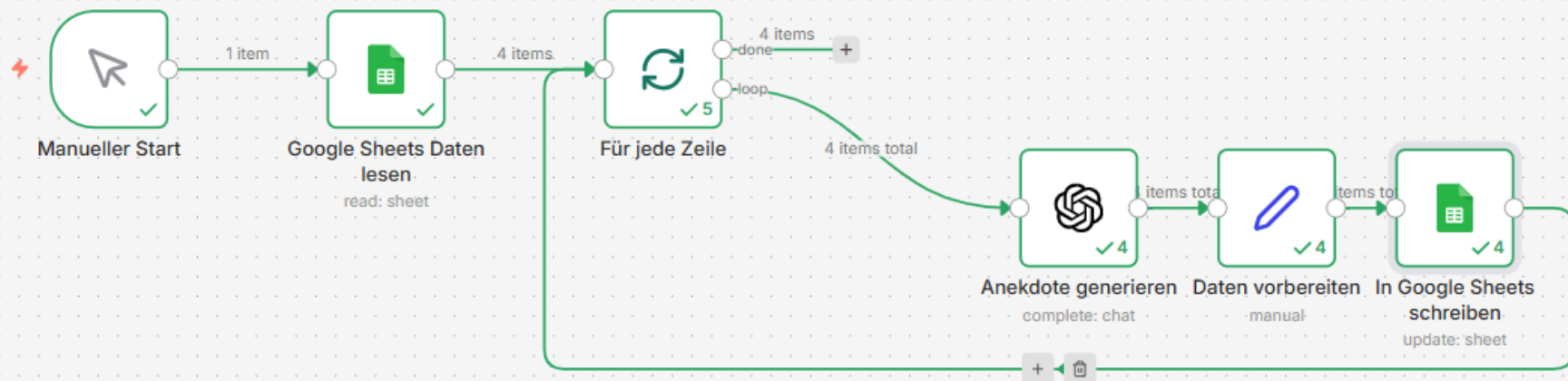
Format News Item:

- Wraps in HTML for email
- Prepares for aggregation

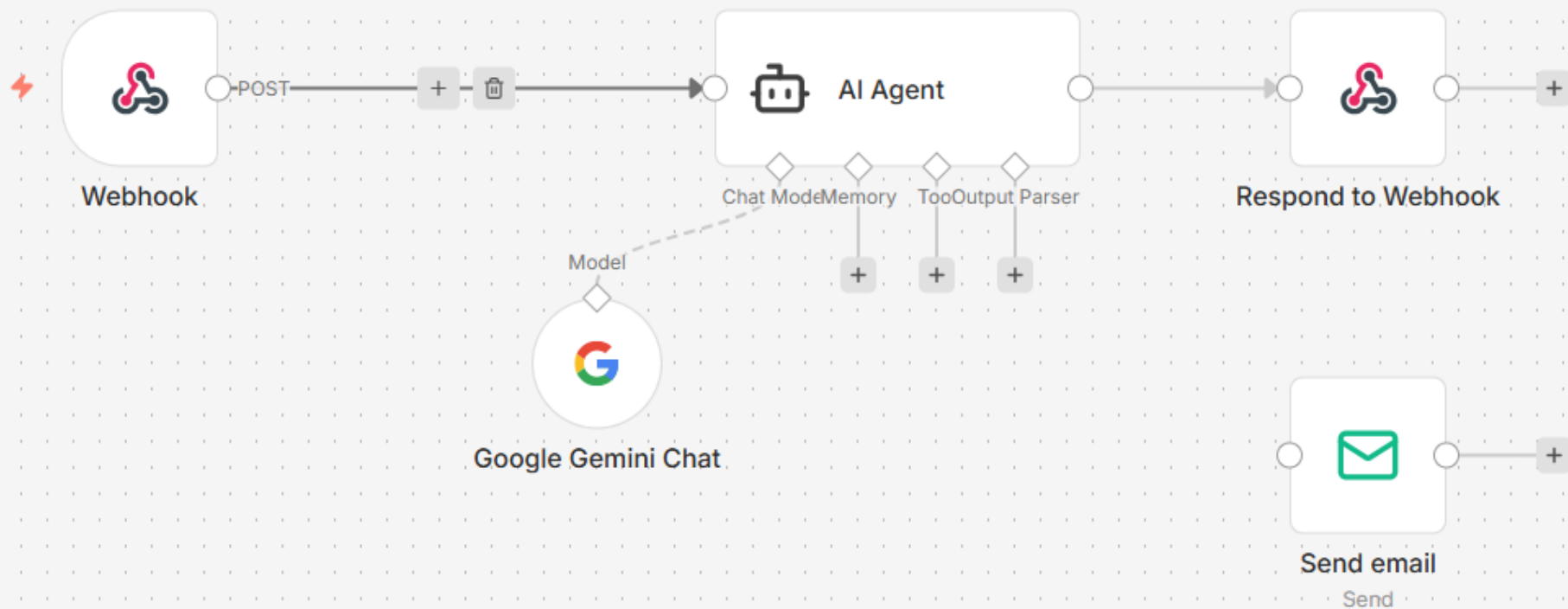
Combine All Summaries:

- Merges into single array
- Ready for email template

Workflow with Iterations



Workflow with WebHooks



API-Call

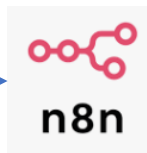
n8n / Webhook Test

POST

Docs Params Authorization Headers (8) Body Scripts Tests Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL

```
1 {  
2   "topic": "Eisschmelze"  
3 }
```



Body Cookies Headers (9) Test Results

```
1  
2 "result": "Greetings, fair traveler. Since thou hast beckoned a bard to speak of the weeping frost, I pray thee attend  
3 to these verses:\nBehold how Phoebus, with his golden eye,\nDoth pierce the mantle of the ancient deep;\nThe crystal  
4 towers that once did touch the sky,\nIn liquid sorrow now begin to weep.\n\nTheir icy crowns, by winter's hand long  
5 wrought,\nDissolve to tears beneath the warming breath;\nWhat once was firm is now to nothing brought,\nA watery change  
6 that mimics slow-paced death.\n\nAs silver bergs surrender to the main,\nThe world doth mourn its crown of frozen  
7 light;\nNo frost may bind these liquid bonds again,\nWhen day's fierce heat o'ercomes the polar night."  
8
```

200 OK

Workflows via MCP

MCP Server

```
{
  "mcpServers": {
    "n8n-mcp": {
      "command": "npx",
      "args": ["n8n-mcp"],
      "env": {
        "MCP_MODE": "stdio",
        "LOG_LEVEL": "error",
        "DISABLE_CONSOLE_OUTPUT": "true",
        "N8N_API_URL": "https://your-n8n-instance.com",
        "N8N_API_KEY": "your-api-key"
      }
    }
  }
}
```

Source: <https://github.com/czlonkowski/n8n-mcp>

MCP Client

Einstellungen - Entwickler

Windows-MCP

Airtable MCP Server

airbnb

n8n-mcp

Windows-MCP running

Dieser Server wird von einer Erweiterung verwaltet

Befehl

uv

Argumente

```
--directory C:\Users\BertGollnick\AppData\Roaming\Claude\Claude
Extensions\ant.dir.cursortouch.windows-mcp run main.py
```

```
"n8n-mcp": {
  "command": "npx",
  "args": ["n8n-mcp"],
  "env": {
    "MCP_MODE": "stdio",
    "LOG_LEVEL": "error",
    "DISABLE_CONSOLE_OUTPUT": "true",
    "N8N_API_URL": "https://n8n.srv1263894.hstgr.cloud/",
    "N8N_API_KEY": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.e"
  }
}
```

MCP Server

To add this MCP to Claude Desktop, update your `claude_desktop_config.json`:

```
{
  "mcpServers": {
    "n8n-workflows Docs": {
      "command": "npx",
      "args": [
        "mcp-remote",
        "https://gitmcp.io/Zie619/n8n-workflows/tree/main"
      ]
    }
  }
}
```

MCP Client

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